

1 THE RECOGNITION THEOREM

1.1 Window syzygies decouple sheet by sheet: the census is q -free

Scope (honest, up front): the theorem below is proven in full for every tower level and every coefficient degree $e < q$. It makes the window syzygy census a FIXED, q -independent, finite linear-algebra problem. It does not by itself close Ley 1; it is the solving tool for the census. Pending the P0 cold gate (Ley 45).

1.1.1 0. Setting

$S = F[s \dots s]$, $E = (e, e, e) = \bigcup I_J$ radical (Odd Symmetric, pillar i), the 15 sheets L_J (matchings of K), $R = S/E$, $q = 3^v$. For a **pair** $\{a, b\} \in \{0 \dots 5\}$ let $V(\{a, b\}) := \bigcup I_J \{a, b\}$ L_J (the three sheets whose matching contains the edge) and $I(\{a, b\}) := I(V(\{a, b\}))$.

Lemma 0 (the edge ideal). $I(\{a, b\}) = (s_a + s_b) + (\sigma_k(ab), \sigma_{q-k}(ab))$, where $\sigma_k(ab)$ is the k -th elementary symmetric polynomial of the four variables other than s_a, s_b . *Proof.* Every sheet through the edge satisfies $s_a + s_b = 0$, and its restriction to that hyperplane is a matching-sheet of the K on the remaining four variables; the union of the three K -sheets has radical ideal $(\sigma_k(ab), \sigma_{q-k}(ab))$ — the K instance of the Odd Symmetric Theorem (the intersection of the K matching ideals is generated by the odd elementary symmetric polynomials). Adding the hyperplane form gives the ideal of the union.

1.1.2 1. The Recognition Theorem

Theorem R. Let $q = 3^v$ and let $\phi = (\phi_1, \dots, \phi_5)$ be homogeneous of degree $e < q$. Then $\sum_{i=0}^5 \phi_i s_i^q \in E$ $\iff \phi_a - \phi_b \in I(\{a, b\})$ **for every one of the 15 edges $\{a, b\}$ of K** . Moreover the right-hand condition does not involve q : **for $e < q$, the space Syz_e of window syzygies is canonically independent of the tower level.**

Proof. $F := \sum \phi_i s_i^q \in E$ $\iff F$ vanishes on every sheet (E radical). Fix a sheet J with pairs (a_k, b_k) ; on L_J , $s_{b_k} = -s_{a_k}$, and with sheet coordinates $u_k := s_{a_k}|_{L_J}$: $F|_{L_J} = \sum_{k=1}^3 (\phi_{a_k} - \phi_{b_k})|_{L_J} \cdot u_k^q$ (using $s_{b_k}^q = -u_k^q$, q odd). Each summand is $u_k^q \cdot g_k$ with $\deg g_k = e < q$; two monomials $u_k^q \cdot m$ and $u_l^q \cdot m$ with $k \neq l$ can never coincide (each would need exponent q in two different slots while the cofactors have degree $< q$). Hence the three summands have disjoint monomial supports, and $F|_{L_J} = 0 \iff (\phi_{a_k} - \phi_{b_k})|_{L_J} = 0$ for each k . Collecting the condition over the three sheets through a fixed edge $\{a, b\}$ gives exactly $\phi_a - \phi_b \in I(\{a, b\})$ (Lemma 0). The conditions are linear constraints on degree- e forms with no reference to q .

Remark (sharpness). The decoupling uses only $e < q$; at $e = q$ (degree $2q$, the window's top edge) monomial collisions $u_k^q u_l^q$ become possible and the statement is exactly where the Zone-C boundary analysis (Theorem Z and the $d = 2q$ residue) takes over. The theorem covers the entire open window $q \leq d \leq 2q-1$.

1.1.3 2. Corollaries

Corollary 1 (q-free census). For each $e \geq 0$ the dimension $_e := \dim \text{Syz}_e/(\text{trivial})$ is a fixed integer, and for every $q > e$: $\dim (\mathfrak{m}^{\{[q]\}}R)_{\{q+e\}} = 5 \cdot \text{HF}(R)_{_e} - _e$. **The graded Hilbert function of the gap ideal throughout the window is independent of the tower level.** The tower has vanished from the census side of the wall.

Corollary 2 (the solving tool). Finding all window syzygies at coefficient degree e is the finite linear system: $_a - _b \cdot (s_{_a+s_b}) + (\wedge^{\{ab\}}, \wedge^{\{ab\}})$ over the 15 edges — 15 membership conditions in fixed ideals. In R the $_$ -slack collapses ($\wedge^{\{ab\}} - (s_{_a+s_b}) \bmod E$): **the only genuinely new slack is $\wedge^{\{ab\}}$** — every syzygy beyond the even/known families must spend $_$ -slack on some edge.

anchors (byte-verified). $e = 0$: $_ = 0$ (5 raw generators measured). $e = 1$: $_ = 0$. $e = 2$: $_ = 1$ = the $p_{\{q+2\}}$ syzygy (trivial S -type) ($74 = 75-1$ at $v=2$). $e = 3$: $_ = 5 = p_{\{q+2\}} \cdot R$ (standard type) ($165 = 170-5$). $e = 4$: $_ = 25$, decomposing (evidence: $\text{dims} + \text{mod-3 trace}$) as $\text{triv triv std}(4,2) (4,2)$ — known 16 + the extracted 9 (explicit machine specimens in hand; closed-form naming open).

1.1.4 3. Ley 41 + status

Objects fresh: the edge-decoupling, Lemma 0's K reduction as used here, the q -freeness. Distinguished: the §0 H -Koszul alias concerns the pair ideal on Λ (different object); §234 S -averaging (not used — S only organizes). **Status: Theorem R + Corollaries = (b)-grade v , pencil, with anchors; pending P0. The remaining open content of P2: name the $(4,2)$ generator at $e = 4$ (specimens extracted), establish the census families for all e (with the q -freeness, a single fixed sequence), then assembly.**

Part of the Sofa Theorem repository: github.com/tretoef-estrella · tretoef@gmail.com